

TsaoSciResearcher Engineering Audit

Release 0.5.3 | current-tree evidence | 2026-07-24

Evidence-first scientific research control layer

Question -> evidence -> design -> execution handoff -> validation -> artifact

Truth boundary: a contract or handoff is not an execution receipt.

340

Capability contracts

322

Named catalog contracts

15

Workflows

15

Schemas

25

Test modules

7

Domain packs

Delivered integration

1. Measurement Boundary with calibration, uncertainty and applicability fields
2. Structure-Property planning with confounders, alternatives and scale bridges
3. Bilingual causality guard and evidence-to-source execution-receipt checks
4. Deterministic test/quality HTML-SVG interfaces and this four-page PDF

Repository operation

All changes were written directly to main; no branch and no pull request were created.

Current-tree software validation matrix

Current-tree cross-platform validation

The current full integration run passed the declared repository and engineering gates.

macos python 3 12		PASS
ubuntu python 3 10		PASS
ubuntu python 3 13		PASS
windows python 3 12		PASS

Current-tree validation gates

bandit high severity		PASS
bounded performance		PASS
byte identical release builds		PASS
complete regression		PASS
critical mutation killed		15/15
deterministic visual reports		PASS
json schemas 15		PASS
mypy strict		PASS
repository and contract audit		PASS
reverse order regression		PASS
ruff format and lint		PASS
scientific quality guards		PASS
seeded random order regression		PASS
validation tree digest		PASS

Current-tree PASS records end-to-end repository validation; scientific acceptance remains separate.

Scientific quality controls

Four deterministic contracts: measurement, multiscale reasoning, causal claims and evidence provenance.

Examples: PASS 3 | WARN 0 | BLOCK 1 | average completeness 89%

Measurement boundary **PASS**

Declared completeness: 100%



Operational definition, calibration, uncertainty and applicability are explicit.

MB-COMPLETE: Operational measurement boundary is explicit.

Kind: measurement-boundary

Structure-property planner PASS

Declared completeness: 100%



The chain records intervention, structure, mediator, response and competing...

SP-CHAIN: Structure, mediator and property are all explicit.

Kind: structure-property-plan

Causality guard **BLOCK**

Declared completeness: 57%



Association-only evidence blocks causal wording in English and Chinese.

CG-OVERCLAIM: Causal wording exceeds the declared study design.

Kind: causality-guard

Evidence traceability	PASS
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Declared completeness: 100%



Claims link to evidence locators and execution claims require receipts.

ET-COMplete: Claim, evidence, locators and execution boundary are traceable.

Kind: evidence-traceability

BLOCK is intentional when wording exceeds evidence or an execution claim lacks a receipt.

The guards remain configurable and do not hard-code material-specific trends.

Current-tree validation and reproducibility

Current-tree full integration

GitHub Actions run: 30029705214

Repository, regression, static, mutation, performance and release gates passed.

Validated source tree

SHA-256: 220c2f2a73a39aebce15106ff3647be32f4e093eeb641b2b38e21aa799003c46

Validated files: 300

Permanent generated artifacts

Test dashboard, scientific-quality dashboard, PDF and checksum: PASS

Critical mutations killed: 15/15

Reproduction entry points

```
python scripts/audit_repository.py
```

```
python -m pytest -q -p hypothesis.extra.pytestplugin
```

```
python scripts/build_test_dashboard.py --check
```

```
python scripts/build_research_quality_dashboard.py --check
```

```
python scripts/build_engineering_report.py --check
```

```
python scripts/generate_checksums.py --check
```

Scientific boundary

Software PASS validates the declared repository controls and source-tree digest.

It does not certify external experiments, calculations, legal conclusions or safety decisions.

All changes were published directly to main; no branch and no pull request were created.